

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 02

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

- A** 6
- B** -6
- C** -12
- D** -3
- E** 3
- F** -2

2 Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

A $2x^2 + x - 3$

B $2x^2 - 5x + 9$

C $2x^2 - x + 3$

D $2x^2 - x - 3$

E $2x^2 + x + 3$

F $2x^2 - x + 5$

- 3** A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?
- A** at least 1, at most 5
 - B** exactly 5
 - C** at least 1, at most 4
 - D** at least 0, at most 4
 - E** at least 0, at most 5
 - F** at least 2, at most 5

- 4** An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I** If $d > 0$ then every term after some point is positive.
- II** The sum of the first n terms is a quadratic function of n .
- III** If a and d are integers then S_n is an integer for every n .

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 5 In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?
- A P is necessary and sufficient for Q.
 - B P is necessary but not sufficient for Q.
 - C P is sufficient but not necessary for Q.
 - D P is neither necessary nor sufficient for Q.

- 6** A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 7 A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?
- A** 6
- B** 16
- C** 10
- D** 8
- E** 4
- F** 12

- 8** The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum.
What is that sum?

- A** 45
- B** 15
- C** 20
- D** 18
- E** 12
- F** 10

- 9** The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?
- A** 4
 - B** 3
 - C** 10
 - D** 25
 - E** 15
 - F** 5

- 10** Consider the statement about a real number x : **if** $x^2 = x$, **then** $x = 1$. Which one of the following is true?
- A** The statement is true and its converse is true.
 - B** The statement is true and its converse is false.
 - C** The statement is false and its converse is true.
 - D** The statement is false and its converse is false.

- 11** A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2 \dots$ ' At which point does the argument break down compared with the $\sqrt{2}$ proof?
- A** $\sqrt{4}$ cannot be written as p/q .
 - B** $p^2 = 4q^2$ does not imply that p is even.
 - C** $k^2 = q^2$ does not show that q is even, so no contradiction arises.
 - D** It is not valid to assume p and q have no common factor.
 - E** The argument is valid, so $\sqrt{4}$ is irrational.

12 The function f has $f'(x) = (x - 1)(x - 3)^2$. Which of the following statements is/are true?

- I** f has a local minimum at $x = 1$.
- II** f has a local maximum at $x = 3$.
- III** f is increasing for all $x > 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 13** For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?
- A** The premises are inconsistent unless n is a multiple of 4 but not of 6; in particular n cannot be a multiple of 12.
- B** n is not a multiple of 4.
- C** n is a multiple of 12.
- D** n is a multiple of 6.
- E** Nothing can be deduced.

14 Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I** It is not raining, or the ground is wet.
- II** If the ground is not wet, then it is not raining.
- III** If the ground is wet, then it is raining.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

15 Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

A $f(x) = x^3 - 3x$ at $a = 1$

B $f(x) = x$ at $a = 0$

C $f(x) = x^2$ at $a = 0$

D $f(x) = x^3$ at $a = 0$

E $f(x) = x^4$ at $a = 0$

16 The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I** If $ac < 0$, the equation has two distinct real roots.
- II** If $b = 0$, the equation has no real roots.
- III** If the equation has a repeated root, then $b^2 = 4ac$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

17 A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I** The angle opposite c is obtuse.
- II** c is the longest side.
- III** The triangle cannot be isosceles.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

18 The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?

- I** The equation $f(x) = g(x)$ has exactly three real solutions.
- II** The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
- III** $f(x) - g(x)$ is a polynomial of degree at least 3.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

19 Which of the following statements is/are true?

- I** For every real number x , there exists a real number y such that $y > x$.
- II** There exists a real number y such that for every real number x , $y > x$.
- III** For every real number x , there exists a real number y such that $xy = 1$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

20 Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I** S contains a positive multiple of 5.
- II** S contains no positive multiple of 3.
- III** S contains a negative number.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

END OF TEST